

Software Requirements Specification (SRS)

Group-03

Name	ID
Afsana Hena	2252421109
Marjia Khatun	23524202071
Mihir Bormon	23524202121
Alvee Ahnaf Khan	23524202141

Title: Food Waste Management System: Connecting Restaurants with Leftovers to Orphanages

1. Introduction

1.1 Purpose

This document sets out the full software requirements for the Food Waste Management System. We're building this as a Java-based web application to link restaurants with surplus food to orphanages that could use it. The SRS covers all key functions, limitations, and objectives, helping everyone involved developers, managers and stakeholders get on the same page before development kicks off.

1.2 Intended Audience

This is aimed at:

- **Project Managers:** To track progress and keep things aligned with business objectives.
- **Software Developers (Java):** For clear technical and functional specs to guide implementation.
- **Quality Assurance (QA) Engineers:** To build test plans and check that the software meets requirements.
- **System Architects:** To plan the overall architecture and database design.

- **Stakeholders (Restaurant Owners, Orphanage Managers):** To review features and give approval.

1.3 Intended Use

We'll refer to this SRS across the entire development lifecycle:

- As the starting point for system architecture and UI design.
- To direct developers during coding and rollout.
- For testers as a checklist to confirm requirements are covered.
- As a reference for handling new features or updates down the line.

1.4 Product Scope

The Food Waste Management System is a fresh web platform focused on cutting food waste and tackling hunger through real-time donations.

In Scope: Registration for restaurants and orphanages, posting food donations, searching listings, claiming items, and built-in notifications/messaging for coordinating pickups.

Out of Scope: Handling physical delivery, payments, or monitoring food temperature in route.

Benefits: Helps restaurants avoid tossing good food, gives orphanages a steady meal supply, and keeps a clear log of contributions for transparency.

1.5 Risk Definitions

- **System Downtime:** Server crashes mean restaurants can't post listings and orphanages miss out leading to wasted food that could have been saved.
- **Data Privacy Violation:** Breaches exposing phone numbers, addresses, or locations could spark legal trouble and break trust.
- **Food Safety Incident:** Claimed food that has expired or mishandled might cause health issues, hurting the platform's credibility.
- **Impersonation Risk:** Fake restaurants might fish for orphanage details or fake orphanage may try to scam the restaurant.

2. Overall Description

2.1 User Classes and Characteristics

- **Restaurant Partner:** Usually a manager or chef remains quite busy during rush hours. They want a straightforward and optimized interface for fast listings with as little typing as possible.

- **Orphanage Partner:** The orphanage admin or cook who handles the food department of it. They rely on up-to-date details like availability, amounts, and pickup windows.
- **System Administrator:** Admin that handles the account approvals, platform monitoring, dispute fixes, and reports on impact.

2.2 User Needs

- **Restaurant Need:** An easy tool to alert nearby orphanages about leftover food that requires as little time as possible.
- **Orphanage Need:** One-stop spot for donation alerts, with solid information on what's offered, how much, and when to collect for better meal planning.
- **Common Need:** A simple communication platform that allowss basic discussion, like "Use the back door" or "Bring your own containers."

2.3 Operating Environment

Server-Side: Runs on cloud or on-prem Linux/Windows servers, needing Java Runtime Environment (JRE) and MySQL database.

Client-Side: Accessible through common browsers (Chrome, Firefox, Safari, Edge etc) on desktops, laptops, tablets, or phones. The app has to be fully responsive for all screens.

2.4 Constraints

Technical Constraints: Backend in Java; frontend with HTML, CSS, JavaScript. The database must be relational SQL (It may change later).

Regulatory Constraints: Full compliance with local privacy laws on handling personal data.

2.5 Assumptions

- We assume restaurants only list safe, properly handled surplus food.
- Orphanages can manage their own transport.
- All users have basic web skills and a suitable browser-equipped device.

3. Requirements

3.1 Functional Requirements

- The system shall allow new users to register as either a "Restaurant" or an "Orphanage."
- The system shall collect name, address, email, phone number, and password during registration.

- The system shall verify the user's email address before activating the account.
- The system shall allow registered users to log in securely using their email and password.
- The system shall provide an authenticated restaurant user with a form to post a new donation.
- The form must include fields for: Food Item Name, Quantity, Pickup Deadline and any Special Instructions.
- Upon submission, the system shall save the donation with a status of "Available".
- The system shall display a list of all "Available" donations to authenticated orphanage users.
- The list shall display the restaurant name, food item, quantity, and pickup deadline for each donation.
- The system shall allow orphanages to sort the list by Pickup Deadline or Date Posted.
- The system shall then display the contact details (phone, address) of the orphanage to the restaurant and vice versa.
- The system shall allow a restaurant user to mark a claimed donation as "Picked Up" once the transaction is complete.
- The system shall move completed donations to a history log for both the restaurant and the orphanage.

3.2 Non-Functional Requirements

- The system shall support at least 100 simultaneous users during peak hours, and the main donation dashboard shall load in less than 3 seconds for users on a standard broadband connection.
- All user passwords shall be hashed and salted using a secure algorithm before storage, and all data transmitted between the user's browser and the server shall be encrypted.
- The system shall be protected against common web vulnerabilities, including SQL Injection
- In the event of a crash, the system shall automatically restart and recover to its last known good state.
- The user interface shall be intuitive and require minimal training. A new user should be able to post or claim a donation within 2 minutes of their first login. The application shall be fully responsive and functional on mobile devices with small screen sizes
- The system shall display a mandatory liability waiver during the registration process, which users must accept. Additionally, a disclaimer shall appear on every donation listing stating that the food quality and safety are the sole responsibility of the donating restaurant.